



Online Shoppers Purchasing Intention

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Abstract

This study aims to predict online shopper purchasing intention using a dataset of 12,330 user sessions described by 18 behavioral and demographic variables sourced from Google Analytics. The target variable, Revenue, indicates whether a given session resulted in a purchase. Due to the significant class imbalance present in the dataset, the Synthetic Minority Over-sampling Technique (SMOTE) is applied to ensure balanced model training. Principal Component Analysis (PCA) is conducted for exploratory analysis and dimensionality reduction to identify the most informative features driving purchase behavior. Six classification models are then built and evaluated: Logistic Regression, Decision Tree, Random Forest, XGBoost, K-Nearest Neighbors (KNN), and a PCA-based Logistic Regression. Model performance is assessed using accuracy, precision, recall, and F1-score on a 70/30 train-test split. The goal is to determine which model most effectively distinguishes buyers from non-buyers and to translate the findings into actionable insights for e-commerce conversion strategies. Results indicate that ensemble methods, particularly Random Forest and XGBoost, achieved the strongest predictive performance. These models outperformed the simpler approaches due to their ability to capture non-linear relationships and complex feature interactions within the data, which a standard linear model, such as Logistic Regression, is unable to account for. These findings provide practical insights for improving e-commerce targeting and conversion optimization.

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Introduction

E-commerce has grown rapidly in recent years. It becomes more and more important to understand why some visitors make purchases and some don't. Research has studied the impact of technology, trust of websites, and convenience of online shopping as major determinants of consumer behavior. Other studies have shown that shipping options and costs are also a big factor in a person's decision to buy. These factors became even more important with the COVID-19 pandemic, which led more consumers to shop online.

Accordingly, the study of consumer behavior has long been a core topic within the discipline of marketing. Thereby, researchers have often used behavioral economics and psychological theories to better understand how consumers make decisions. These approaches help companies understand what motivates consumers to buy, how they respond emotionally, and what they think about, and also help in providing a strong foundation for developing effective marketing strategies.

In recent years, the rapid growth of digital technology has significantly changed consumer behavior, shifting many activities from offline environments to online platforms. This transformation has created both new opportunities and challenges for researchers and businesses. Through social media, mobile applications, and e-commerce platforms, companies now have access to large amounts of consumer behavior data. This data has become a valuable component in understanding and predicting consumer behavior more accurately.

Literature Review

The research in [1] and [2] looked at the problem of purchase intention prediction as a supervised learning task and utilised historical data collected from an online bookshop to classify user sessions into two classes, i.e., browsing sessions and buyer sessions. The two studies have used different machine learning approaches for this classification. The work of [1] has investigated Support Vector Machines (SVMs) with different kernel functions, whereas [2] has used the k-Nearest Neighbour (k-NN) algorithm.

In an earlier work, Mobasher et al. [4] evaluated two clustering approaches applied to user transaction and pageview data to find meaningful group profiles. The profiles were created to help recommendation systems provide effective personalisation to visitors in the early stages of browsing an online store.

Building on this, Moe [5] established the framework for a system that can perform particular actions based on the type of visit. In order to achieve this, the researcher developed a system that can classify clicks on online stores into four categories: purchasing, browsing, searching, and learning. The approach entails analysing in-store browsing behaviour, particularly the features of the pages clicked on, and classifying visits using a k-means clustering technique.

Data Description

The dataset comprises 18 variables spanning three categories: behavioral metrics derived from Google Analytics, session-level attributes, and user demographic proxies. The table below describes all variables:

Variable	Type	Description
Administrative	Integer	Number of administrative pages visited
Administrative_Duration	Float	Total time spent on administrative pages (seconds)
Informational	Integer	Number of informational pages visited
Informational_Duration	Float	Total time spent on informational pages (seconds)
ProductRelated	Integer	Number of product-related pages visited
ProductRelated_Duration	Float	Total time spent on product pages (seconds)
BounceRates	Float	Google Analytics metric measuring the percentage of visitors who leave a site after viewing only one page, without interacting further
ExitRates	Float	Google Analytics of the percentage of visitors who leave a site from a specific page after visiting one or more pages in a session
PageValues	Float	Google Analytics that calculates the average monetary value a specific page contributes to a session
SpecialDay	Float	Closeness of visit to a special day (0–1 scale)
Month	Categorical	Month of the session (Feb–Dec)
OperatingSystems	Integer	Encoded OS of the visitor's device
Browser	Integer	Encoded browser type used
Region	Integer	Encoded geographic region of the visitor
TrafficType	Integer	Encoded traffic source type

VisitorType	Categorical	New_Visitor, Returning_Visitor, or Other
Weekend	Boolean	Whether the session occurred on a weekend
Revenue	Boolean	Whether the session resulted in a purchase

Figure 1: Variables, data types, and description

Objective

To identify the primary drivers of online shopper conversions, this project employs a suite of classification models, including Logistic Regression, Random Forest, XGBoost, KNN, Naive Bayes, and Decision Trees, to predict the binary target variable, Revenue. By evaluating these methods through key performance metrics such as accuracy, precision, recall, and the F1-score, we can determine which model most effectively distinguishes buyers from non-buyers. While ensemble methods like Random Forest and XGBoost often provide superior predictive accuracy, the inclusion of Decision Trees allows for the extraction of straightforward "if-then" rules that explain the specific browsing patterns leading to a purchase. Ultimately, this comparative approach serves to translate complex data patterns into actionable insights that can be applied to real-world e-commerce strategies.

Statistical Methods Used

- 1. Principal Component Analysis (PCA):** Used for exploratory data analysis and dimensionality reduction. It transforms the original variables into a new set of uncorrelated variables called Principal Components (PCs) that capture the maximum variance in the dataset.
- 2. PVE (Proportion of Variance Explained):** Calculated using eigenvalues to determine how much information each Principal Component captures. We use Scree Plot to decide how many components are necessary to represent the data accurately.
- 3. Logistic Regression:** A baseline probabilistic model used for binary classification. It models the log-odds of the target variable Revenue as a linear combination of the predictor variables.
- 4. K-Nearest Neighbors (KNN):** A non-parametric, instance-based learning algorithm that classifies observations based on the majority class of their "k" closest neighbors in the multi-dimensional feature space.

5. **Naive Bayes:** A classification technique based on Bayes' Theorem with an assumption of independence among predictors.
6. **Random Forest:** Random Forests are a collection of decision trees built using bootstrap aggregation (bagging). By averaging the predictions gained from many decorrelated trees, they reduce variance and improve stability over using individual trees [3].
7. **XGBoost (Extreme Gradient Boosting):** XGBoost is a high-performance ensemble learning algorithm that functions by constructing a series of decision trees in a back-to-back fashion. In this process, every subsequent tree is specifically designed to minimize the residual errors left behind by the ones before it. It is highly optimized for performance and speed [3].
8. **Synthetic Minority Over-sampling Technique (SMOTE):** SMOTE is a statistical method for balancing the amount of cases in your dataset. From pre-existing minority cases that you provide as input, the component creates new instances [6].

Data Cleaning

The dataset consists of 12,330 data points across 18 columns. Upon inspection, no missing values were detected across any of the variables, meaning no imputation or row removal was necessary, allowing us to work with the full dataset.

Regarding data types, the Boolean fields Revenue and Weekend were confirmed as binary and converted to integer format with values of 0 and 1. The categorical variables Month and VisitorType required encoding before being used in modelling, and were handled using one-hot encoding to create separate binary columns for each category level.

Finally, feature scaling was identified as a necessary step for certain variables. Continuous variables such as BounceRates, ExitRates, and PageValues span very different numerical ranges, which can negatively affect distance-based algorithms like KNN that are sensitive to input scale. To address this, z-score standardisation was applied to these variables.

Data Transformation

Upon examining the raw data, several transformations were identified as necessary before modelling could begin. The duration and count features were found to be heavily right-skewed, so a log transformation using `log1p` was applied to `Administrative_Duration`, `Informational_Duration`, `ProductRelated_Duration`, and `ProductRelated` to compress the long right tail and make these features more suitable as model inputs.

For the categorical variables, one-hot encoding was applied to `Month` and `VisitorType`, which have 10 and 3 categories, respectively, converting each category into a separate binary column so the models can interpret categorical information without imposing any artificial ordering between levels.

Accordingly, the Boolean columns `Revenue` and `Weekend` were cast from their original `true/false` format to binary integer values of 0 and 1, ensuring these fields are treated as numeric inputs across all models used in the project.

Data Analysis

1. Exploration of Data

Our main target variable was `Revenue`, this would help us explore whether a particular browsing session actually led to a purchase.

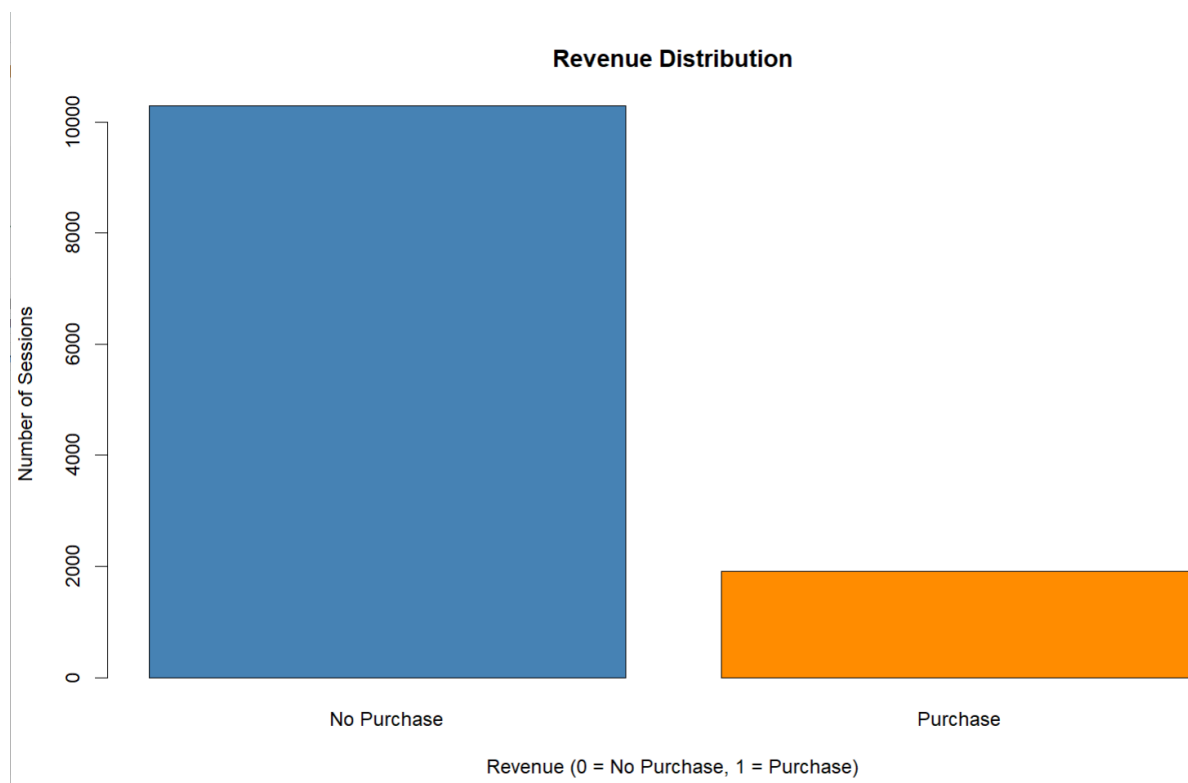


Figure 2: Revenue Distribution

The observation we noticed from this was the severe class imbalance. The "No Purchase" bar reaches 10,422 sessions, while the "Purchase" bar sits at around 1,908 sessions. This means roughly 84% of all sessions did not result in a purchase, while only about 16% did. This imbalance has direct implications for our project.

Thus, we implemented a Synthetic Minority Over-sampling Technique (SMOTE) on the dataset to improve model performance on underrepresented data.

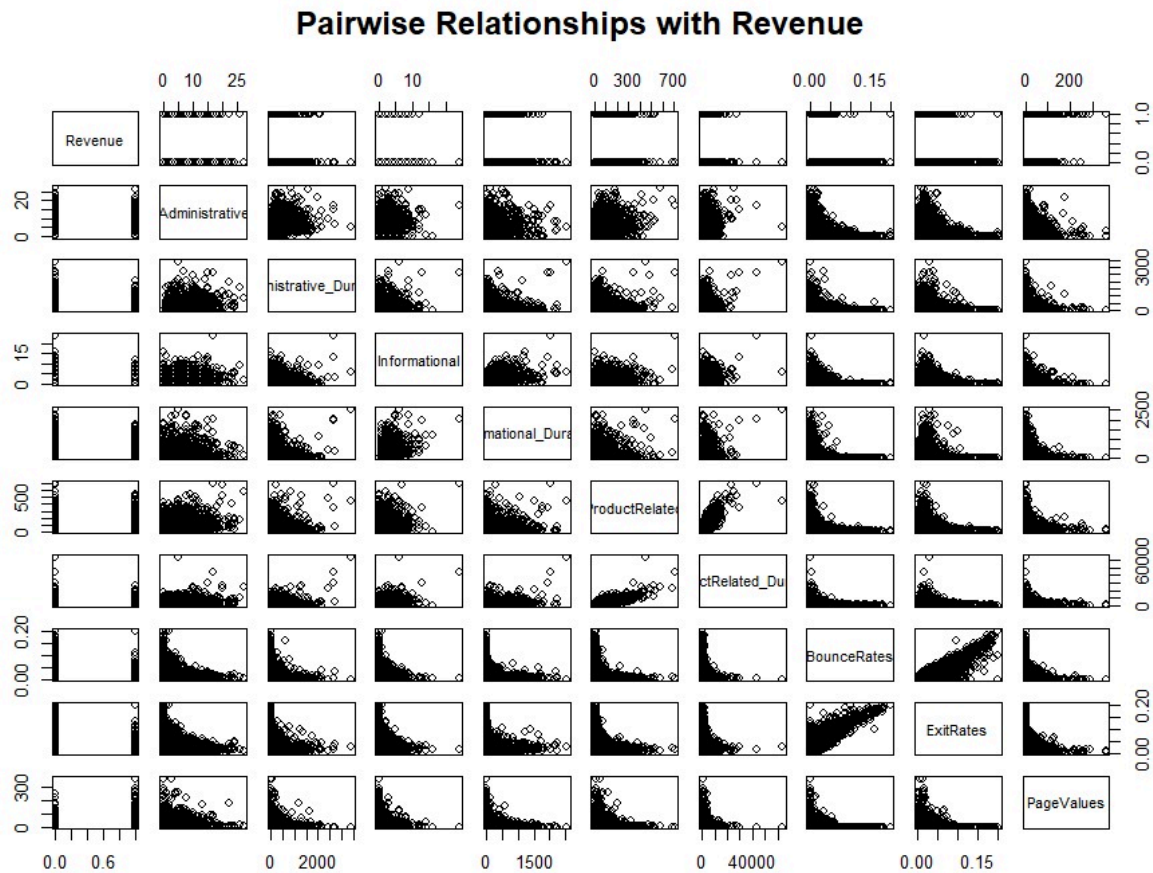


Figure 3: Pairwise Relationship with Revenue

From this plot, we observe that BounceRates and ExitRates are strongly positively correlated (+0.9). This is expected since both measure how quickly visitors leave the site; a visitor who bounces will also have a high exit rate.

There is a clear positive relationship between ProductRelated and ProductRelated_Duration (+0.86). The more product pages a visitor views, the more time they spend on them.

The same pattern holds for the administrative pages and their duration (+0.6). This is expected and just confirms the variables are internally consistent.

Overall, almost every variable pair shows the same bottom-left clustering pattern, meaning most sessions are low engagement across the board.

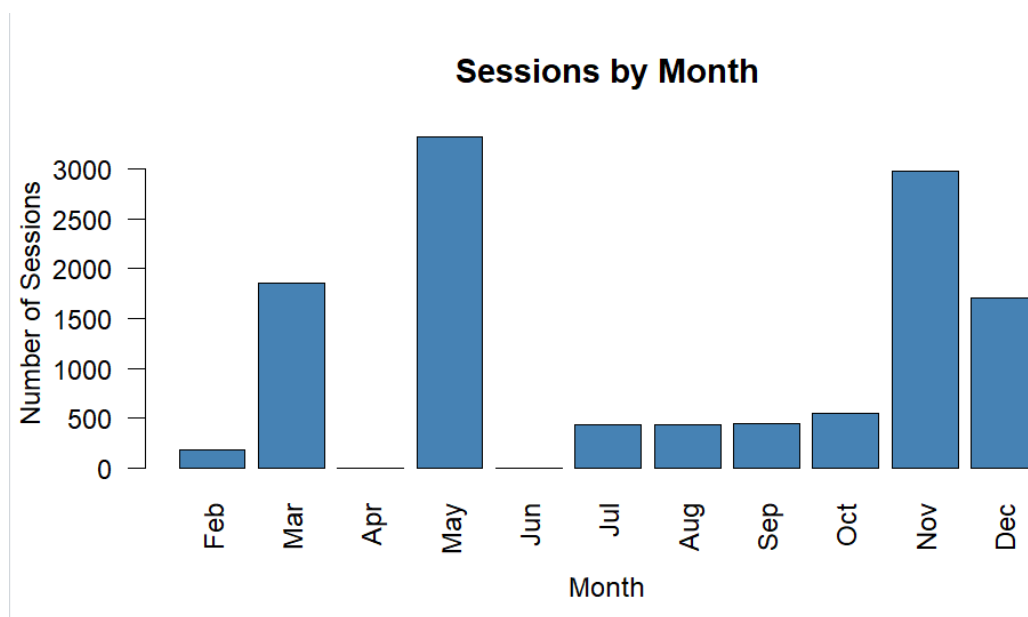
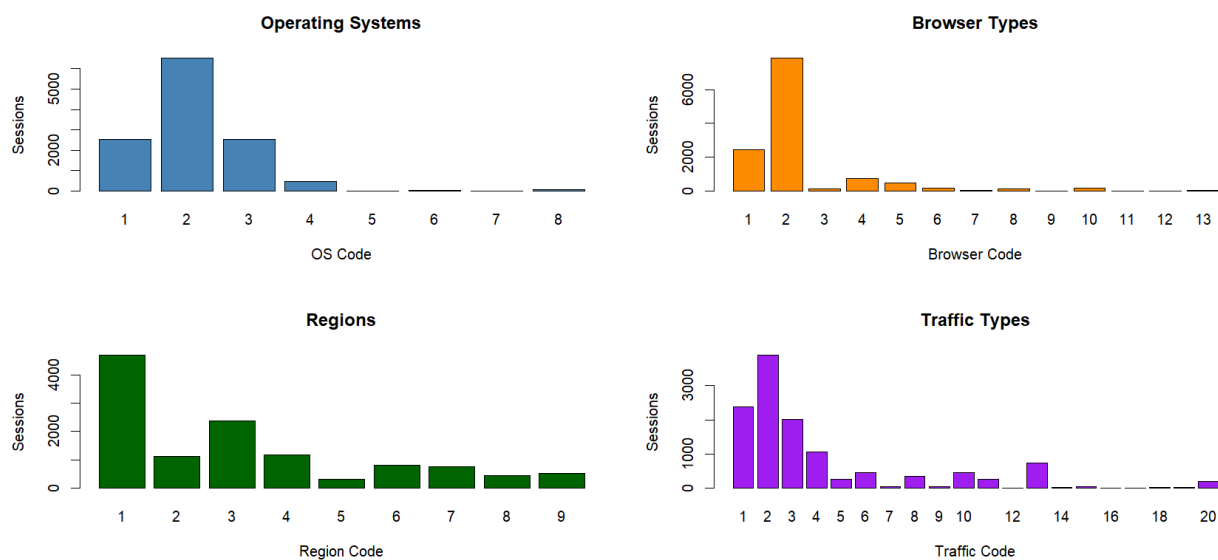


Figure 4: Session Distribution by Month



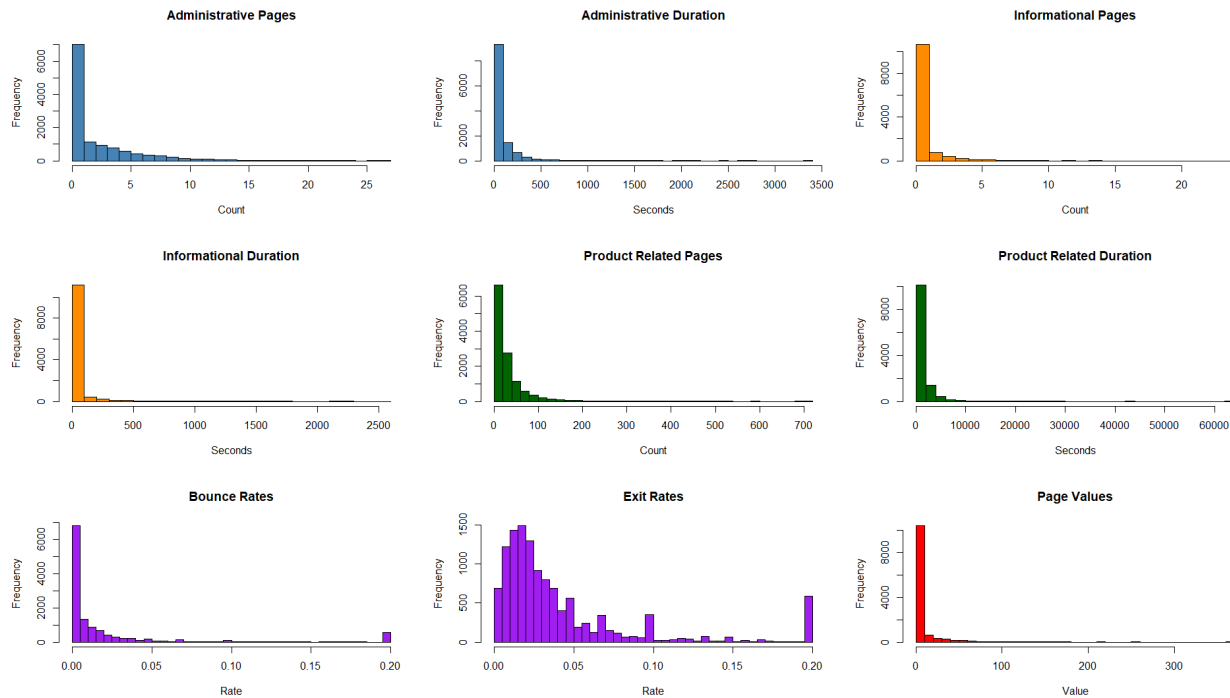


Figure 5: Plots for Descriptive Features

Based on the plots, the data reflects user session behavior on a website, characterized by heavily right-skewed distributions across nearly all metrics. Most users interact with a very small number of pages (Administrative, Informational, and Product Related) for short durations, with a significant majority of sessions resulting in low page values and exit/bounce rates clustered near zero.

2. Feature Selection using PCA

Principal Component Analysis (PCA) is utilized to capture the maximum variance within a dataset using a minimized set of principal components. By converting high-dimensional data into uncorrelated variables, this transformation maintains the important information. The final scores provide a useful basis for further statistical analysis, even though the components themselves might not be directly interpretable.

```
=== Eigenvalues (Variance per PC) ===
> print(round(eigenvalues, 4))
[1] 3.3599 1.8058 1.4398 1.0904 1.0361 0.9609 0.9568 0.8755 0.8362 0.7426
[11] 0.3842 0.3273 0.0994 0.0851
```

Figure 6: Eigenvalues of Principal Components

Kaiser-Guttman Criterion: The decision would be to take components with an eigenvalue ≥ 1 which explains more variance than a single observed variable. In this case, there are 5 components greater than 1. PC1, PC2, PC3, PC4, and PC5 all show eigenvalues greater than 1.

```
> summary(pr.out)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	1.833	1.344	1.1999	1.04421	1.01790	0.98023	0.97814
Proportion of Variance	0.240	0.129	0.1028	0.07788	0.07401	0.06863	0.06834
Cumulative Proportion	0.240	0.369	0.4718	0.54971	0.62372	0.69235	0.76069

	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.93566	0.91446	0.86175	0.61985	0.57212	0.3153	0.29167
Proportion of Variance	0.06253	0.05973	0.05304	0.02744	0.02338	0.0071	0.00608
Cumulative Proportion	0.82322	0.88295	0.93600	0.96344	0.98682	0.9939	1.00000

Figure 7: Cumulative Proportion of Principal Components

Analysis of the Chosen PCs: Based on the Eigenvalues, the first 6 PCs account for approximately 70% of the variance.

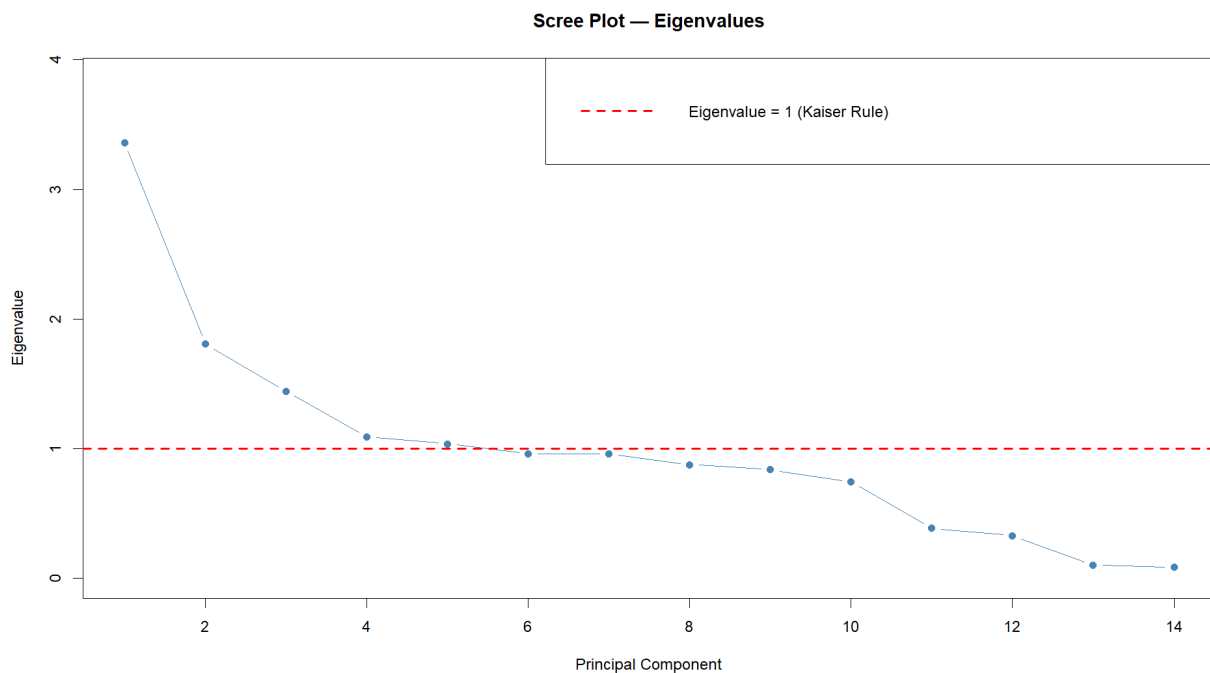


Figure 8: Scree Plot

The scree plot result shows that there appears to be an elbow at 5. Which means PC1 through PC5 should be used for analysis.

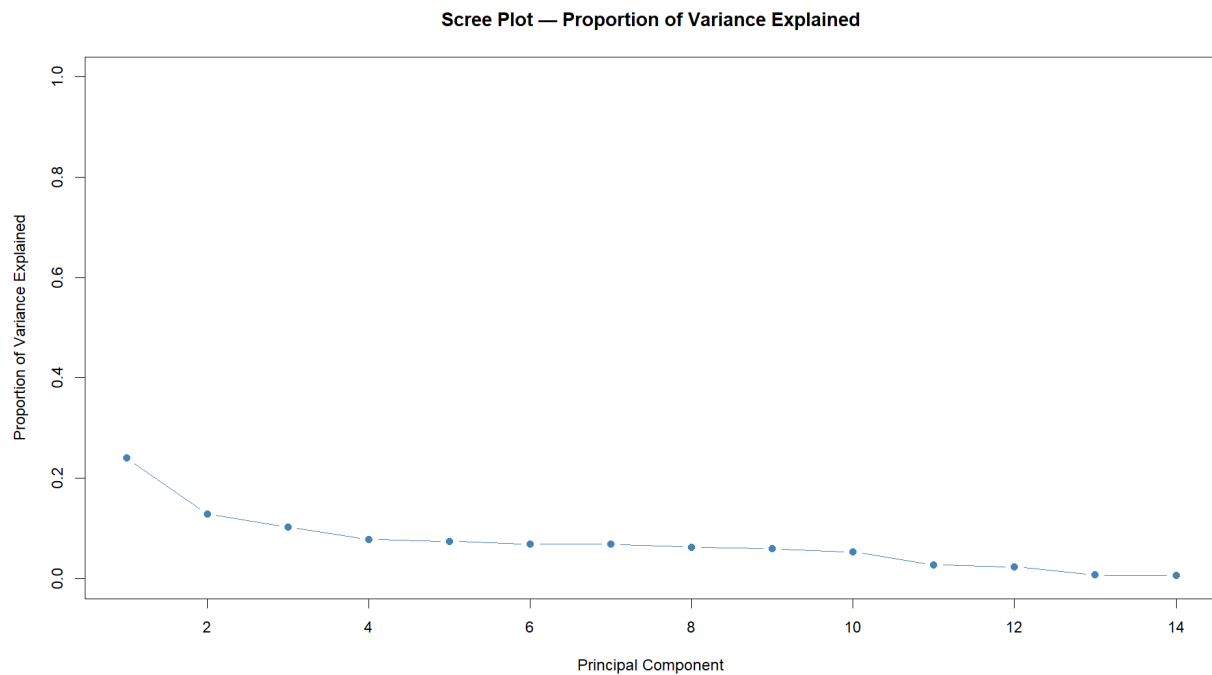


Figure 9: Scree Plot for Proportion of Variance

Similarly, through the proportion of variance plot, PC1 accounts for approximately 24% of the total variance, followed by PC2 at around 13% and PC3 at roughly 10%. After this point, the curve flattens out significantly, with all remaining components contributing very little individually. The elbow of the plot appears around PC3 to PC4, suggesting that the first 3 to 4 components capture the most meaningful structure in the data.

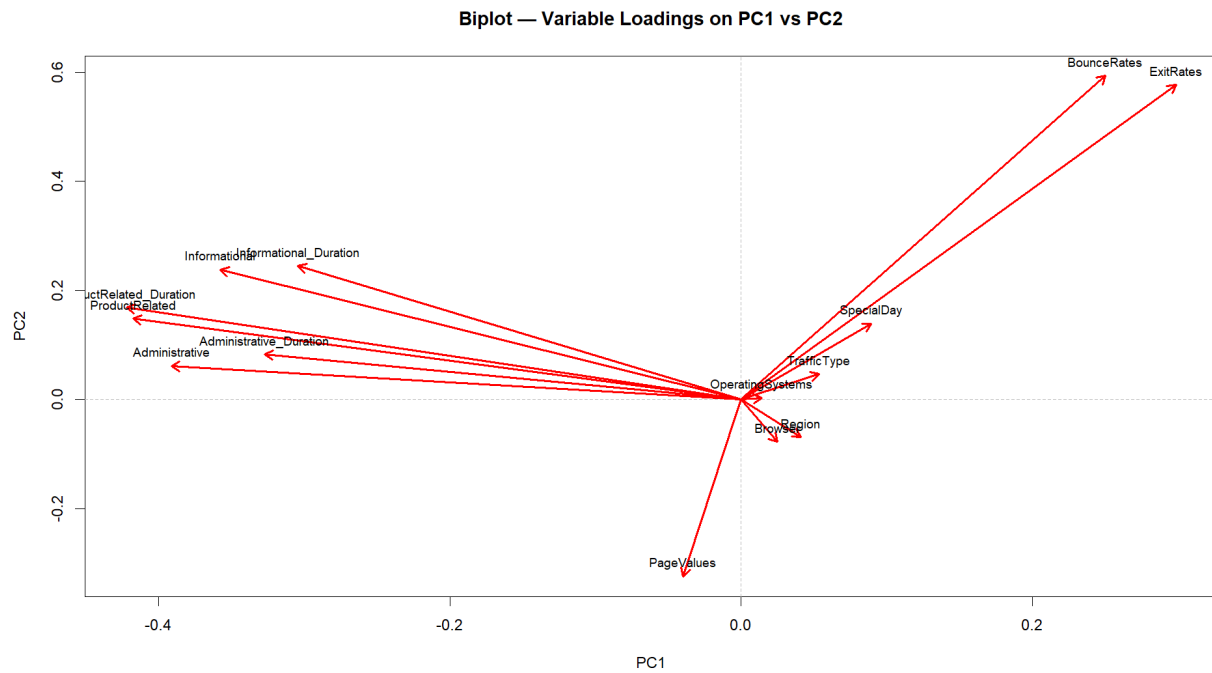


Figure 10: Biplot for PC1 and PC2

From the biplot, we can interpret that:

- The browsing activity variables (Administrative, ProductRelated, Informational, and their durations) all point strongly to the left, meaning they are the main drivers of PC1 and are positively correlated with each other.
- BounceRates and ExitRates point to the upper right, opposite to the browsing variables, confirming their strong correlation with each other and their inverse relationship with browsing engagement.
- PageValues points sharply downward, loading independently on PC2, reinforcing its role as a distinct and important predictor separate from the other variables.
- Region, Browser, and OperatingSystem have very short arrows, indicating they contribute little to either component and carry less predictive information.

```

=== Loadings: PC1 and PC2 ===
> print(round(pr.out$rotation[, 1:2], 4))
              PC1      PC2
Administrative -0.3913  0.0616
Administrative_Duration -0.3272  0.0826
Informational -0.3577  0.2383
Informational_Duration -0.3045  0.2450
ProductRelated -0.4177  0.1482
ProductRelated_Duration -0.4219  0.1684
BounceRates  0.2502  0.5934
ExitRates    0.2986  0.5768
PageValues   -0.0400 -0.3242
SpecialDay    0.0893  0.1391
OperatingSystems 0.0143  0.0031
Browser       0.0247 -0.0777
Region        0.0410 -0.0688
TrafficType    0.0535  0.0469
> |

```

Figure 11: Loadings for PC1 and PC2

From the loadings, we can tell that:

- ProductRelated_Duration (-0.42) and ProductRelated (-0.42) are the strongest drivers of PC1, representing overall browsing engagement.
- Administrative (-0.39), Informational (-0.36), and their durations also load strongly on PC1, confirming it captures general browsing activity.
- BounceRates (0.59) and ExitRates (0.58) dominate PC2, meaning this component specifically captures how quickly visitors leave without purchasing.

From these observations, we can infer that PC1 separates interested buyers from disinterested ones, which is directly relevant to predicting purchases. While PC2 isolates bounce and exit behaviour alongside PageValues are two of the strongest known indicators of purchase intent.

Overall, the PCA analysis provided a clearer understanding of which variables carry the most information in this dataset. Rather than replacing the original features, these findings serve as a foundation for the modelling phase, giving us greater confidence in the variables we are working with. The identification of PageValues, BounceRates, ExitRates, and the ProductRelated variables as the most informative features suggests that these will likely play a significant role in our classification and regression models.

3. Model Building

As mentioned previously, SMOTE was conducted on the dataset after cleaning and transformation to solve the issues related to imbalance. Having done SMOTE, the proportion of the Revenue distribution has become very similar.

0	1
0.5190805	0.4809195

Figure 12: Revenue Distribution after SMOTE

The resulting target variable has an approximate proportion of 50/50. For the classification analysis, we conducted five methods, including Logistic Regression, Random Forest, XGBoost, KNN, and Naive Bayes. To incorporate our previous analysis, we performed a separate PCA Regression to observe any significant changes.

The model analysis phase is designed to evaluate and compare multiple classification techniques to determine the most effective approach for predicting visitor purchase intent. For the purpose of our model, we have used a 70/30 training/test data partition.

```
> cat("Train size:", nrow(train_df), "\n")
Train size: 13886
> cat("Test size :", nrow(test_df), "\n")
Test size : 5951
```

Figure 13: Training and Testing size

Because Random Forest and XGBoost can handle high-dimensional interactions and non-linear correlations, they were used as the main high-performance models. While XGBoost was used for its gradient boosting framework, which iteratively corrects faults from earlier iterations, Random Forest was chosen for its robust bagging technique, which lowers variance by averaging several decision trees. These techniques were given priority in order to capture the complex behaviors of online shoppers, such as the compounding effect of low ExitRates and high PageValues.

To establish a linear baseline for how well the classes can be separated using a traditional additive technique, logistic regression was utilized. To enhance this, Naive Bayes was utilized to evaluate a feature-independent probabilistic method. This was particularly useful in determining whether specific category factors, such as Month or VisitorType, had considerable independent predictive weight.

To categorize sessions, the system uses K-Nearest Neighbors (KNN). By using this method, the system compares a session to previous sessions that have similar features according to their distance. It is the reason that decision trees were chosen for the model. For those models, the

structure is easy to follow. In this way, the trees show how the system makes a prediction when it follows logic that is direct and uncomplicated, "if-then".

Principal Component Analysis (PCA) was implemented as a feature selection and data summarization tool to determine if the 18 original variables could be condensed into a smaller set of uncorrelated components without losing predictive integrity.

4. Model Results

4.1. Logistic Regression

For the Logistic Regression, we initially made the model based on all the variables to get a baseline estimate of the dataset after SMOTE. Our target variable, being Revenue, we observed many variables having significance related to it. Predominantly, the ones we proved through PCA analysis. PageValues, ExitRates, BounceRates, and Month_Nov have high estimates.

Coefficients: (2 not defined because of singularities)				
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-8.179e-01	2.393e-01	-3.418	0.000630 ***
Administrative	-2.156e-03	1.162e-02	-0.186	0.852792
Administrative_Duration	-5.663e-04	1.788e-04	-3.168	0.001536 **
Informational	2.999e-03	3.074e-02	0.098	0.922275
Informational_Duration	-2.278e-04	2.132e-04	-1.069	0.285157
ProductRelated	-3.079e-03	1.510e-03	-2.039	0.041405 *
ProductRelated_Duration	1.639e-04	3.204e-05	5.116	3.13e-07 ***
BounceRates	-5.934e+00	2.277e+00	-2.606	0.009151 **
ExitRates	-1.035e+01	2.060e+00	-5.026	5.02e-07 ***
PageValues	1.396e-01	3.230e-03	43.234	< 2e-16 ***
SpecialDay	-1.136e-03	1.811e-01	-0.006	0.994995
OperatingSystems	-2.549e-02	3.016e-02	-0.845	0.398009
Browser	3.592e-02	1.562e-02	2.299	0.021489 *
Region	-1.457e-02	1.059e-02	-1.376	0.168834
TrafficType	1.955e-04	6.750e-03	0.029	0.976892
Weekend	1.155e-01	5.808e-02	1.989	0.046671 *
log_Administrative_Duration	5.998e-02	1.794e-02	3.343	0.000828 ***
log_Informational_Duration	4.102e-02	2.389e-02	1.717	0.085922 .
log_ProductRelated_Duration	-1.881e-01	4.573e-02	-4.113	3.91e-05 ***
log_ProductRelated	3.553e-01	7.875e-02	4.512	6.42e-06 ***
Month_Aug	-1.798e-02	1.700e-01	-0.106	0.915748
Month_Dec	-6.860e-01	1.354e-01	-5.067	4.04e-07 ***
Month_Feb	-1.919e+00	4.460e-01	-4.302	1.69e-05 ***
Month_Jul	-9.104e-02	1.713e-01	-0.531	0.595170
Month_June	-2.347e-01	2.105e-01	-1.115	0.264919
Month_Mar	-8.092e-01	1.352e-01	-5.986	2.15e-09 ***
Month_May	-1.077e+00	1.334e-01	-8.075	6.76e-16 ***
Month_Nov	6.284e-01	1.236e-01	5.084	3.70e-07 ***
Month_Oct	-1.782e-01	1.567e-01	-1.137	0.255572
Month_Sep	NA	NA	NA	NA
VisitorType_New_Visitor	4.038e-01	7.539e-02	5.356	8.53e-08 ***
VisitorType_Other	-3.185e-01	4.251e-01	-0.749	0.453784
VisitorType_Returning_Visitor	NA	NA	NA	NA

Figure 14: Logistic regression

Variables like Informational, SpecialDay, and OperatingSystems have high p-values (greater than 0.05). This means they don't have a statistically significant impact on predicting Revenue in this specific model.

```

Confusion Matrix and Statistics

      Reference
Prediction 0    1
0  2734  632
1   355 2230

      Accuracy : 0.8341
      95% CI   : (0.8244, 0.8435)
    No Information Rate : 0.5191
    P-Value [Acc > NIR] : < 2.2e-16

      Kappa : 0.6666

  McNemar's Test P-Value : < 2.2e-16

    Sensitivity : 0.8851
    Specificity : 0.7792
   Pos Pred Value : 0.8122
   Neg Pred Value : 0.8627
    Prevalence : 0.5191
    Detection Rate : 0.4594
    Detection Prevalence : 0.5656
    Balanced Accuracy : 0.8321

    'Positive' Class : 0

```

Figure 15: Result of Logistic Regression

The biggest issue here is the False Negatives (632). This is reflected in the Specificity (0.7792). Identifying only 77% of buyers is quite low for an e-commerce model, as you're effectively "missing" nearly a quarter of your revenue-generating sessions.

4.2. Decision Tree

We performed a cross-validation in order to obtain the best value of size for the pruned tree. The best size being 8 from the eventual deviance obtained.

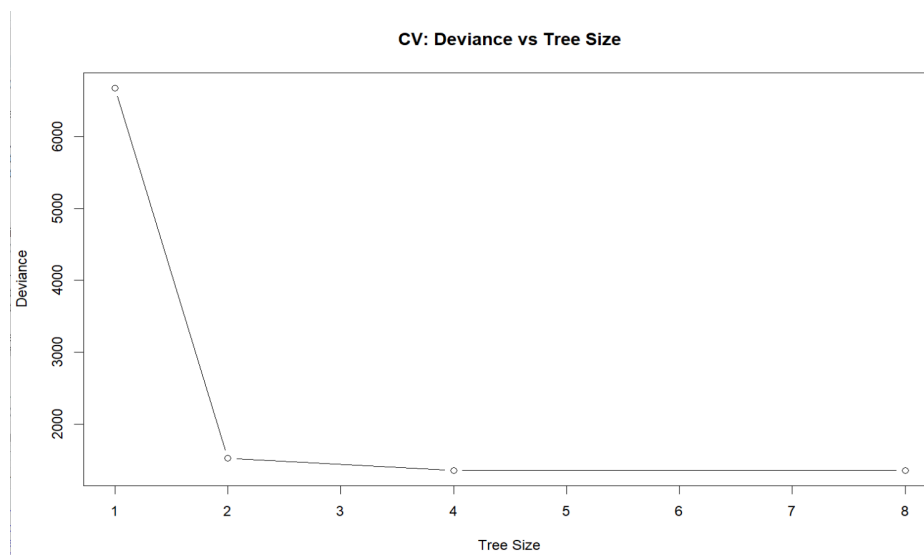


Figure 16: Best-size plot for Decision Tree

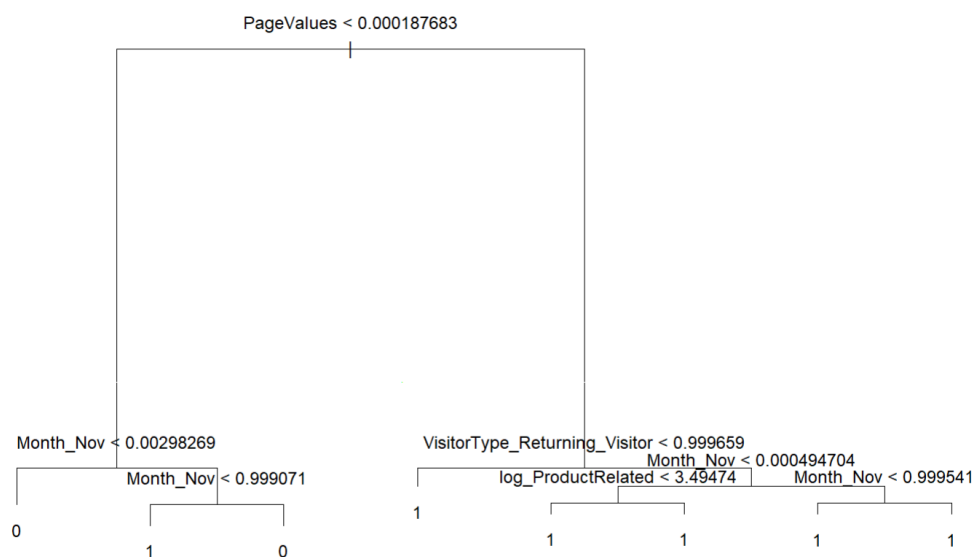


Figure 17: Decision Tree Model

According to the tree, the most important component in forecasting a sale is PageValues. It is the best "separator" in the data because it is at the top. The user is taken to the right side of the tree, where nearly every path leads to a "1" (a purchase) if a session has a high Page Value. This implies that consumers who engage with valuable material are pre-sold on purchases.

The model further refines the paths using browsing intensity metrics like `log_ProductRelated`, demonstrating that once a high-value page is identified in a returning visitor, the depth of the visitor's product engagement becomes a decisive factor in the final classification.

Confusion Matrix and Statistics

```

                Reference
Prediction      0      1
0      2608    247
1       338   2615

Accuracy : 0.8993
95% CI : (0.8912, 0.9069)
No Information Rate : 0.5072
P-Value [Acc > NIR] : < 2.2e-16

Kappa : 0.7986

McNemar's Test P-Value : 0.0001984

Sensitivity : 0.8853
Specificity : 0.9137
Pos Pred Value : 0.9135
Neg Pred Value : 0.8855
Prevalence : 0.5072
Detection Rate : 0.4490
Detection Prevalence : 0.4916
Balanced Accuracy : 0.8995

'Positive' Class : 0

```

Figure 18: Result of Decision Tree Model

By utilizing cross-validation to identify an optimal tree size of 5 nodes, the model successfully minimized deviance while avoiding overfitting, ultimately achieving a solid accuracy of approximately 90% and a balanced F1-score of 0.899.

4.3. Random Forest

The random forest model creates 500 different versions of our training data using the bagging method.

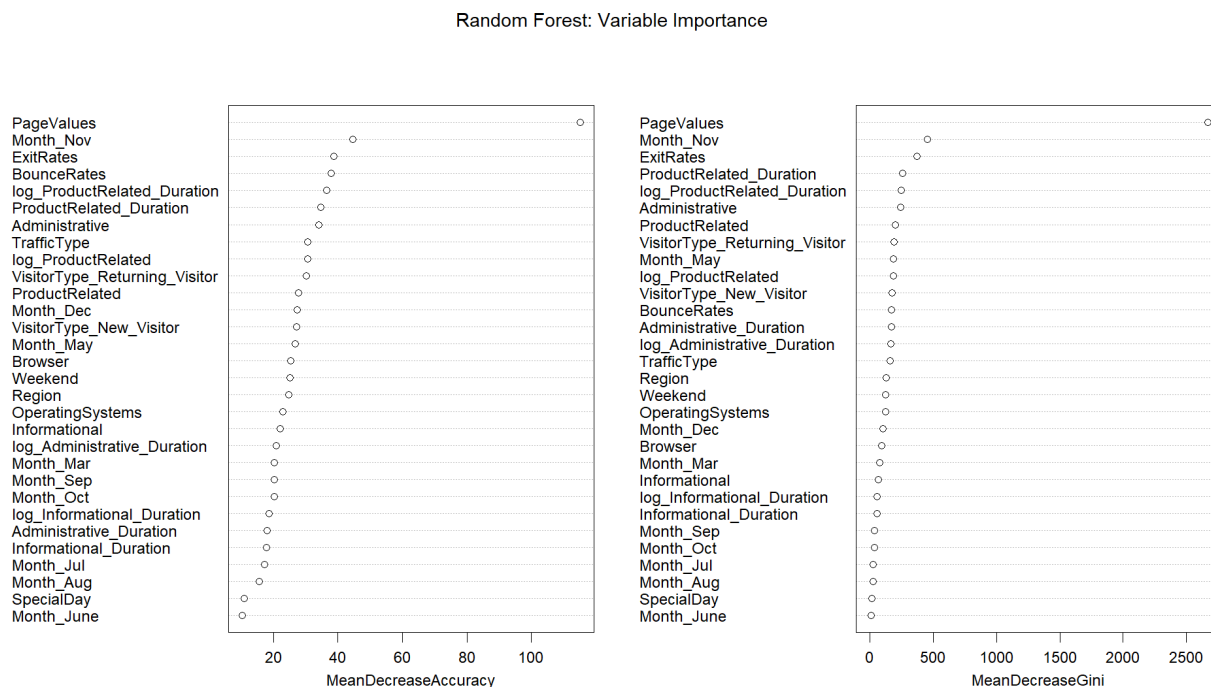


Figure 19: Variable Importance Plot

The Variable Importance Plot confirms that PageValues is the most dominant predictor by a significant margin. In addition, Month_Nov, ExitRates, and ProductRelated_Duration consistently rank as the next influential features, proving that while transaction-related page value is the primary driver, seasonality and user engagement depth are the critical secondary factors in determining purchase intent.

Confusion Matrix and Statistics

	Reference	
Prediction	0	1
0	2893	182
1	196	2680

Accuracy : 0.9365
 95% CI : (0.93, 0.9425)
 No Information Rate : 0.5191
 P-Value [Acc > NIR] : <2e-16

Kappa : 0.8728

Mcnemar's Test P-Value : 0.5037

Sensitivity : 0.9365
 Specificity : 0.9364
 Pos Pred Value : 0.9408
 Neg Pred Value : 0.9318
 Prevalence : 0.5191
 Detection Rate : 0.4861
 Detection Prevalence : 0.5167
 Balanced Accuracy : 0.9365

'Positive' Class : 0

Figure 20: Result of Random Forest Model

By randomly selecting subsets of features for each tree, it prevents a single dominant variable like PageValues from masking the subtle influence of others, such as ExitRates or the November seasonal spike. This was the effect of Bagging, which resulted in a 93.6% Accuracy, making it a good model for handling non-linear complexities.

4.4. XGBoost

XGBoost is more surgical than Random Forest, which constructs deep, independent trees. It constructs trees one at a time, with each new tree concentrating on the misclassifications, based on the previous trees.

```

Confusion Matrix and Statistics

      Reference
Prediction  0    1
0  2907  195
1   182 2667

      Accuracy : 0.9366
      95% CI   : (0.9302, 0.9427)
No Information Rate : 0.5191
P-Value [Acc > NIR] : <2e-16

      Kappa : 0.8731

McNemar's Test P-Value : 0.5366

      Sensitivity : 0.9411
      Specificity : 0.9319
      Pos Pred Value : 0.9371
      Neg Pred Value : 0.9361
      Prevalence : 0.5191
      Detection Rate : 0.4885
      Detection Prevalence : 0.5213
      Balanced Accuracy : 0.9365

      'Positive' Class : 0

```

Figure 21: Result of XGBoost Model

The excellent accuracy indicates that the relationship between PageValues and seasonal variables like Month_Nov was effectively captured by this "boosting" procedure. Having a sensitivity of 94.11%, showcases that it correctly identifies 94% of the users who did *not* make a purchase.

4.5. K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a simple yet effective algorithm that predicts whether a user will make a purchase by looking at the behavior of the k most similar visitors (neighbors) in the dataset.

Resampling: Cross-Validated (10 fold)

Summary of sample sizes: 12497, 12498, 12497, 12499, 12497, 12497, ...

Resampling results across tuning parameters:

k	Accuracy	Kappa
5	0.8754159	0.7509778
7	0.8718878	0.7437870
9	0.8693660	0.7386184
11	0.8673500	0.7345237
13	0.8634614	0.7266705
15	0.8629564	0.7256092
17	0.8623803	0.7244301
19	0.8609408	0.7215542
21	0.8585640	0.7167633
23	0.8584916	0.7165843
25	0.8575556	0.7146830
27	0.8571234	0.7138022
29	0.8556836	0.7108969
31	0.8551797	0.7098801
33	0.8548193	0.7091328

Figure 22: KNN k-tuning

The model was trained on a 10-fold cross-validation. Having multiple k neighbors, we notice that the Kappa value and accuracy keep decreasing as seen in Figure .

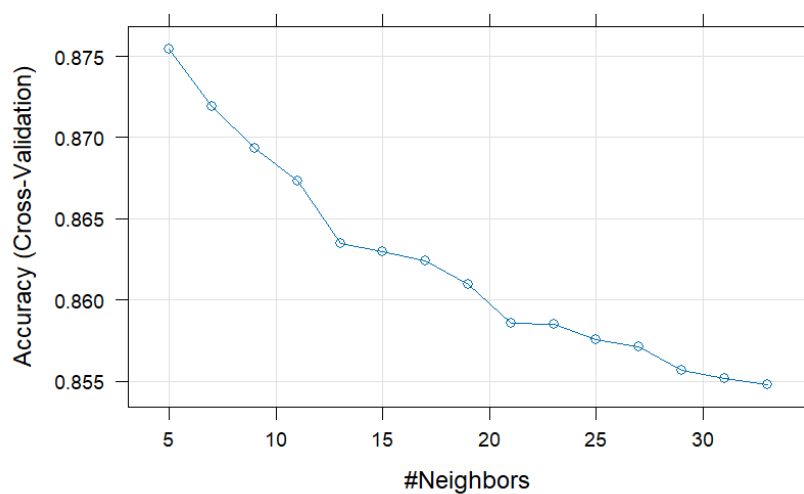


Figure 23: Accuracy plot for K-Neighbours

Thus, from this figure, we pick $K = 5$ to be the optimal value for the model. This confusion matrix in Figure 16 represents the KNN model's performance on the test set. With an Accuracy of 87.15% and a Kappa of 0.7431, the model shows strong, reliable predictive power.

```

                Reference
Prediction    0    1
0  2626  302
1   463 2560

Accuracy : 0.8715
95% CI : (0.8627, 0.8799)
No Information Rate : 0.5191
P-Value [Acc > NIR] : < 2.2e-16

Kappa : 0.7431

McNemar's Test P-Value : 7.259e-09

Sensitivity : 0.8501
Specificity : 0.8945
Pos Pred Value : 0.8969
Neg Pred Value : 0.8468
Prevalence : 0.5191
Detection Rate : 0.4413
Detection Prevalence : 0.4920
Balanced Accuracy : 0.8723

'Positive' Class : 0

```

Figure 24: Result of KNN Model

4.6. PCA Logistic Regression

We wanted to test whether the results obtained from the PCA analysis would benefit the results if they were incorporated into the model. Thus, we performed the Logistic Regression by using the most important features presented in the PCA analysis.

```

Reference
Prediction    0    1
0  2004  866
1   942 1996

Accuracy : 0.6887
95% CI : (0.6766, 0.7006)
No Information Rate : 0.5072
P-Value [Acc > NIR] : < 2e-16

Kappa : 0.3775

McNemar's Test P-Value : 0.07776

Sensitivity : 0.6802
Specificity : 0.6974
Pos Pred Value : 0.6983
Neg Pred Value : 0.6794
Prevalence : 0.5072
Detection Rate : 0.3450
Detection Prevalence : 0.4941
Balanced Accuracy : 0.6888

'Positive' Class : 0

```

Figure 25: Result of PCA Regression Model

While we expect that removing the unnecessary features would improve our model's performance, it made it perform worse. The reason is that e-commerce intent is often driven by very specific, non-linear triggers (like a high PageValue specifically in November), PCA's linear "averaging" of variables is removing the most important signals.

While PCA is great for simplifying large datasets, it is clearly not the right tool for this specific e-commerce dataset. The relationship between user behavior and revenue is too complex for simple linear components to capture efficiently.

5. Model Comparison

	Model	Accuracy	Sensitivity	F1_Score
RF.Accuracy	RF	0.9399105	0.9392396	0.9406765
XGB.Accuracy	XGB	0.9385331	0.9450102	0.9397468
Tree.Accuracy	Tree	0.8992769	0.8852682	0.8991553
KNN.Accuracy	KNN	0.8720730	0.8520027	0.8710741
Logit.Accuracy	Logit	0.8309229	0.8886626	0.8420714
PCA_Logit.Accuracy	PCA_Logit	0.6887052	0.6802444	0.6891334
NB.Accuracy	NB	0.7162534	0.5617787	0.6676079

Figure 26: Model Comparison of Results Table

These results indicate that the Random Forest and XGBoost models were the most effective at identifying potential buyers while minimizing false positives. In contrast, while simpler models like Logistic Regression and Naive Bayes (NB) provided faster computation, they struggled to capture the complex, non-linear patterns of the dataset, with PCA-Logistic performing the poorest at 68.8% accuracy.

Logistic Regression served as a useful baseline, but its lower sensitivity for the purchase class, missing nearly a quarter of all buyers, makes it less ideal for active sales environments where capturing every lead is a priority.

Ultimately, based on its balanced F1-Score of 0.9406, Random Forest is identified as the best model for predicting purchase intent in this study.

Key Findings and Analysis

- **Ensemble models performed the best**

Methods like Random Forest and XGBoost both achieved around 94% accuracy; thus, they can be termed the strongest models in this project. These models were better at handling complex and non-linear relationships in the dataset, especially factors like PageValues and seasonal trends such as Month_Nov.

- **XGBoost had the highest sensitivity**

A sensitivity of over 94% clearly showed that XGBoost was able to correctly identify most sessions where users had purchase intent. This is especially important in an e-commerce setting because it reduces the number of potential customers that the model fails to detect.

- **Decision Tree was easy to interpret**

Although the Decision Tree did not perform as well as the ensemble models, it was used for inference purposes as it was easier to understand compared to more complex models. Its simple "if-then" approach helped show which customer behaviors were the most important in predicting purchases. It confirmed that PageValues was the strongest indicator of purchase intent. It also highlighted factors like the Month_Nov seasonal increase and the number of ProductRelated pages visited. These insights can help businesses better understand customer behavior and improve e-commerce strategies

- **PCA Logistic Regression gave the weakest results**

The reason PCA Logistic Regression performed poorly in our project is that Principal Component Analysis (PCA) is a linear transformation that prioritizes broad patterns over specific, high-impact details. While PCA is effective for simplifying large datasets, it often removes the "surgical" precision required for e-commerce prediction

- **Standard Logistic Regression worked as a decent baseline**

Regular Logistic Regression achieved an accuracy of around 83%, which is reasonable for a baseline model. However, it produced a higher number of false negatives, meaning it missed many sessions that actually resulted in purchases.

These results show how statistics can significantly improve e-commerce strategies. Results from the Random Forest and XGBoost models prove that customer behavior is influenced by combinations of factors and not just simple demographics, along with the understanding of the compounding effect of behaviors, such as how a specific combination of low ExitRates and high PageValues significantly multiplies the likelihood of a sale. XGBoost's high specificity is useful because it helps businesses identify potential customers more accurately and reduce missed opportunities. The Decision Tree provided some useful insights by showing that PageValues had a major impact on purchase intent, suggesting that improving product pages can increase the likelihood of converting visitors into buyers. On the other hand, the weak performance of PCA-based models showed that oversimplifying data can reduce prediction accuracy and miss important customer behavior patterns.

While accuracy was high across ensemble models, the business impact is best understood through the lens of 'Missed Opportunities' (False Negatives). For instance, the Logistic Regression model failed to identify 632 potential buyers. In a real-world scenario, this represents a significant loss of revenue. In contrast, the Random Forest model reduced these missed opportunities to only 182 sessions, making it far more valuable for conversion-focused marketing strategies

Business Application

Maximizing Marketing ROI: Predicting purchase intent helps businesses use their marketing budget better. It allows them to target users who are close to buying with timely discounts or deals. This reduces wasted spending on casual visitors who are unlikely to buy.

Strategic Inventory and Resource Planning: Purchase predictions enable businesses to manage inventory more effectively. They can anticipate demand during busy times like November. This helps avoid stock shortages and supports better pricing strategies.

Enhancing Customer Experience and Retention: Behavioral insights help platforms personalize the shopping experience. They offer product recommendations and improve

customer journeys. They also identify users at risk of leaving without buying, allowing businesses to act and reduce drop-off.

Leveraging Key Behavioral Drivers: PageValues was the strongest predictor of purchase intent, indicating that businesses should prioritize optimizing high-value product and landing pages. High exit and bounce rates signal areas needing improvement, and the higher purchase likelihood among returning visitors suggests that loyalty programs may offer stronger returns than focusing solely on new customer acquisition.

Future Plans

Real-Time Intervention: The models could be deployed in a live setting where purchase intent is predicted as a session is happening. This would allow businesses to instantly respond with personalized offers or prompts at the moment a user is most likely to buy.

Advanced Feature Engineering: The current dataset uses session-level data, but adding more detailed variables such as click patterns, scroll depth, and purchase history could give the models more information to work with and potentially improve their accuracy.

Deep Learning: More advanced models such as recurrent neural networks could be explored to better capture the nature of a user's browsing journey, which traditional classification models do not look into.

Conclusion

This project aimed to predict online shoppers' purchasing intention using a dataset of 12,330 user sessions across 18 variables. After performing the necessary data cleaning, transformation, and applying SMOTE to address the class imbalance present in the target variable Revenue, six classification models were built and evaluated: Logistic Regression, Decision Tree, Random Forest, XGBoost, K-Nearest Neighbors, and PCA-based Logistic Regression.

Across all models, PageValues emerged as the most significant predictor of purchase intent, followed by ExitRates, BounceRates, ProductRelated_Duration, and Month_Nov. This is consistent with the findings of the PCA and variable importance analyses, which identified these features as carrying the most predictive information in the dataset.

In terms of model performance, ensemble methods outperformed the simpler approaches. Random Forest and XGBoost both achieved high accuracy and strong sensitivity, demonstrating their ability to handle the non-linear relationships present in the data. Logistic Regression served as a useful baseline, but its lower sensitivity for the purchase class, missing nearly a quarter of all buyers, makes it less ideal for active sales environments where capturing every lead is a priority. The PCA-based Logistic Regression performed worse than the full-feature models, confirming that compressing the variables into linear components removed important signals that the other models were able to leverage.

Overall, the results show that machine learning models, particularly ensemble methods, are well-suited for predicting online purchasing behavior. The key drivers identified in this study, page value, exit behavior, and seasonal trends, provide actionable insights that e-commerce businesses can use to better understand and target high-intent visitors.

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